

Retail Customer Retention Analytics – Walmart

Submitted By

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Project Overview

- This project is based on Retail Customer Retention Analytics for Walmart. The main purpose of this project is to analyze customer behavior, customer churn, loyalty programs, promotions, and store performance using Power BI.
 - Customer retention is very important for a retail company because retaining old customers is easier and more profitable than acquiring new customers. In this project, different datasets related to customer demographics, transactions, loyalty programs, and churn were analyzed to identify customer trends and business insights.
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- **Problem Statement**
 - Walmart collects a large amount of customer data from stores and online platforms, but the company faces difficulty in understanding:
 - Why customers are churning
 - Which customers are loyal
 - Which regions and channels are underperforming
 - Whether loyalty programs and promotions are effective
 - The objective of this project is to create an interactive Power BI dashboard that provides useful insights regarding customer retention and customer engagement.
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Dataset Information

- The project contains the following datasets:

1. Customer Demographics

- Contains customer details such as:

- *Customer ID*
- *Age*
- *Gender*
- *Region*
- *Income Level*
- *Membership Since*
- *Preferred Channel*

2. Customer Transactions

- *Contains purchase-related information:*
- *Transaction ID*
- *Product Category*
- *Transaction Date*
- *Amount*
- *Promotion Applied*

3. Store Locations

- *Contains store-related details:*
- *Store ID*
- *Store Type*
- *Region*
- *Opening Year*

4. Loyalty Program

- *Contains loyalty details:*
- *Loyalty Tier*
- *Points Earned*
- *Points Redeemed*

5. Churn Labelled Customers

- *Contains churn information:*

- *Last Purchase Date*
 - *Churn Flag*
 - *Churn Reason*
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Task 1: Data Cleaning and Data Modeling

Steps Performed

- *Loaded all datasets into Power BI using Power Query.*
- *Checked for missing values and duplicate records.*
- *Corrected data types for date and numeric columns.*
- *Created relationships between tables using Customer ID and Store ID.*
- *Calculated Columns Created*
- *Membership Duration*
- *This column calculates how long the customer has been associated with Walmart.*
- *Membership_Duration = DATEDIFF(Customer_Demographics[Membership_Since], TODAY(), DAY)*

Transaction Year

- *Extracted year from transaction date.*
- *Transaction_Year = YEAR(Customer_Transactions[Transaction_Date])*

Transaction Month

- *Extracted month name from transaction date.*
- *Transaction_Month = FORMAT(Customer_Transactions[Transaction_Date], "MMM")*

Insights

- *Relationships were successfully created between all tables.*
- *Data cleaning improved data consistency and analysis quality.*
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Task 2: Churn and Retention Metrics

Measures Created

Total Customers

- *Total Customers* = `DISTINCTCOUNT(Customer_Demographics[Customer_ID])`

Churned Customers

- *Churned Customers* =
- `CALCULATE(`
- `COUNTROWS(Churn_Labelled_Customers),`
- `Churn_Labelled_Customers[Churn_Flag] = 1`
- `)`

Churn Rate

- *Churn Rate* = `DIVIDE([Churned Customers], [Total Customers]) * 100`

Visuals Created

- *Churn Rate KPI*
- *Churn by Region*
- *Churn by Channel*
- *Churn by Loyalty Tier*
- *Funnel Chart*

Insights

- *Churn rate is comparatively high in some regions.*
- *Online channels show slightly higher churn compared to store channels.*
- *Higher loyalty tiers show better customer retention.*
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Task 3: Repeat Purchase Analysis

Measures Created

Total Transactions

- *Total Transactions* = `COUNT(Customer_Transactions[Transaction_ID])`

Average Purchase Frequency

- *Avg Purchase Frequency =*
- *DIVIDE(*
- *[Total Transactions],*
- *DISTINCTCOUNT(Customer_Transactions[Customer_ID])*
- *)*

Customer Segmentation

- *Customers were divided into:*
- *Low Tier (0–3 purchases)*
- *Mid Tier (4–8 purchases)*
- *High Tier (9+ purchases)*

Insights

- *Most customers belong to the low purchase segment.*
- *High-tier customers contribute more to repeat purchases.*
- *Some regions show better customer engagement than others.*
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Task 4: Promotion and Loyalty Impact

Measures Created

Promotion Percentage

- *Promo % =*
- *DIVIDE(*
- *CALCULATE(*
- *COUNTROWS(Customer_Transactions),*
- *Customer_Transactions[Promotion_Applied] = "Yes"*
- *),*
- *COUNTROWS(Customer_Transactions)*

) * 100

Average Purchase With Promotion

- Avg With Promo =
- CALCULATE(
• AVERAGE(Customer_Transactions[Amount]),
• Customer_Transactions[Promotion_Applied] = "Yes"
•)

Average Purchase Without Promotion

- Avg Without Promo =
- CALCULATE(
• AVERAGE(Customer_Transactions[Amount]),
• Customer_Transactions[Promotion_Applied] = "No"
•)

Insights

- Customers using promotions spend more on average.
- Higher loyalty tiers have lower churn rates.
- Points redeemed are lower compared to points earned.
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Task 5: Store and Channel Performance

Visuals Created

- Average Transaction Amount by Store Type
- Churn Rate by Store Type
- Opening Year vs Churn Scatter Plot

Insights

- Supercenter stores generate higher average transaction values.
- Online stores show relatively higher churn.

- *Newer stores have slightly lower retention compared to older stores.*
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Task 6: Customer Lifetime Value (CLV) Analysis

Measures Created

CLV

- *CLV =*
- *DIVIDE(*
- *[Total Spent],*
- *[Membership Years],*
- *0*
- *)*

Days Since Last Purchase

- *Days Since Last Purchase =*
- *DATEDIFF(*
- *MAX(Churn_Labelled_Customers[Last_Purchase_Date]),*
- *TODAY(),*
- *DAY*
- *)*

Insights

- *High CLV customers contribute significantly to revenue.*
- *Some high-value customers are inactive and may churn soon.*
- *Customers in premium loyalty tiers generally have higher CLV.*

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Task 7: Final Dashboard and Executive Summary

Dashboard Pages Created

Page 1: KPI Overview

- *Contains:*
- *Total Customers*
- *Churn Rate*
- *CLV*
- *Average Purchase Frequency*
- *Funnel Chart*

Page 2: Loyalty and Promotion Impact

- *Contains:*
- *Churn by Loyalty Tier*
- *Points Earned vs Redeemed*
- *Promotion Analysis*

Page 3: Store and Channel Insights

- *Contains:*
- *Store Performance Analysis*
- *Churn by Store Type*
- *Store Opening Year Analysis*

Page 4: Customer Segmentation

- *Contains:*
- *Customer Segments*
- *Product Preferences*
- *CLV vs Days Since Last Purchase*

Executive Summary

- *The dashboard provides detailed insights into customer retention and customer behavior. The analysis shows that customers in higher loyalty tiers are more engaged and less likely to churn. Promotions positively affect customer spending, but reward redemption is still low. Some store types and channels show higher churn, indicating areas where Walmart can improve customer engagement.*
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Recommendations

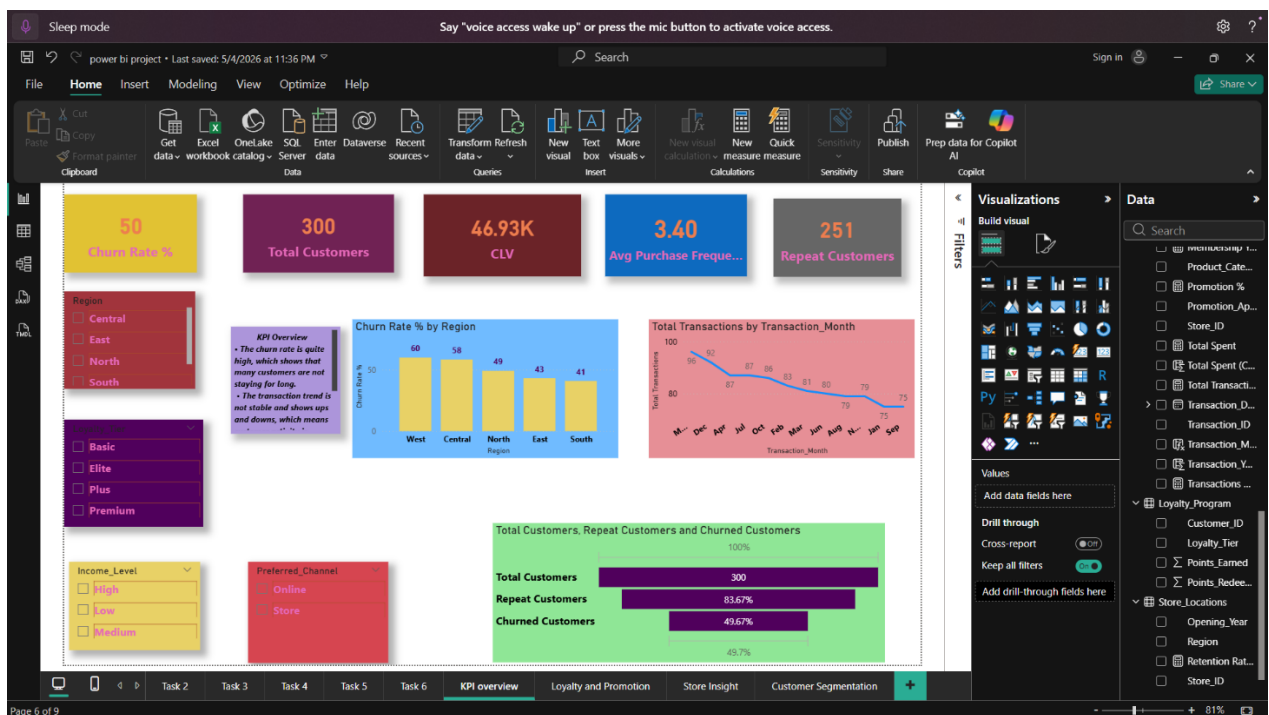
- Focus on high-value customers who are becoming inactive.
- Improve loyalty reward redemption through better offers.
- Strengthen customer engagement in high-churn regions and channels.
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Conclusion

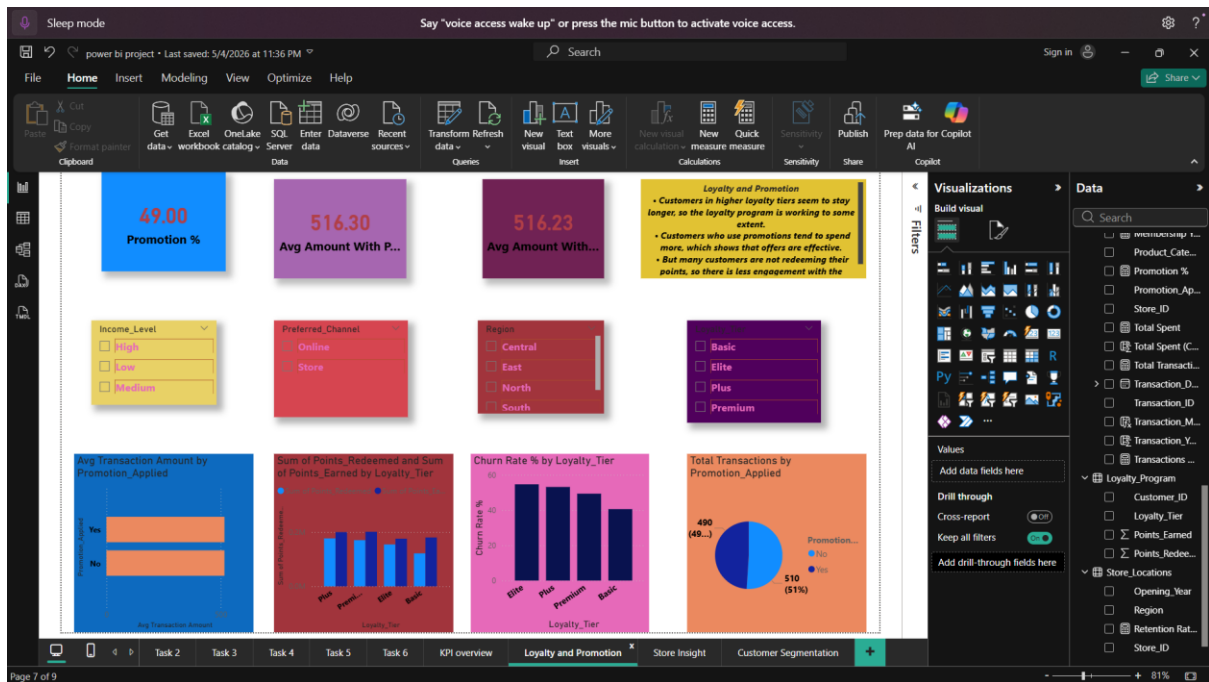
This project helped in understanding customer retention analysis using Power BI. Different visualizations and DAX measures were used to analyze customer behavior, loyalty impact, churn rate, and customer lifetime value. The dashboard can help Walmart improve customer retention and business performance.

Screenshots

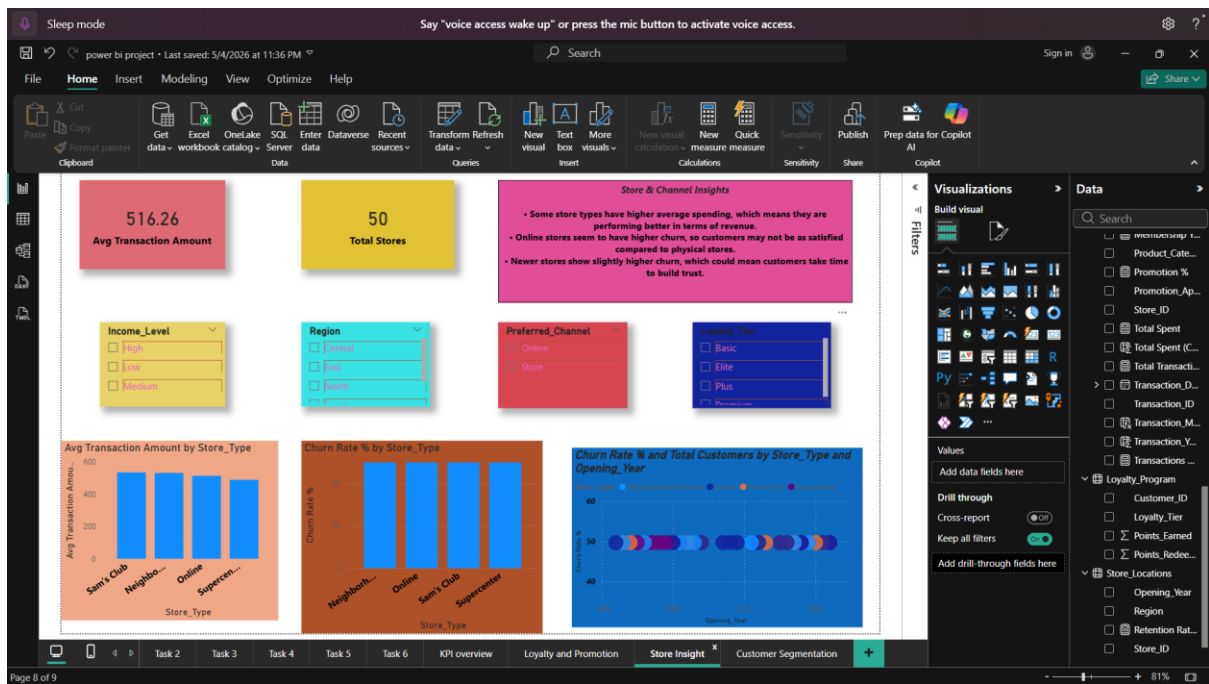
Page 1 Screenshot



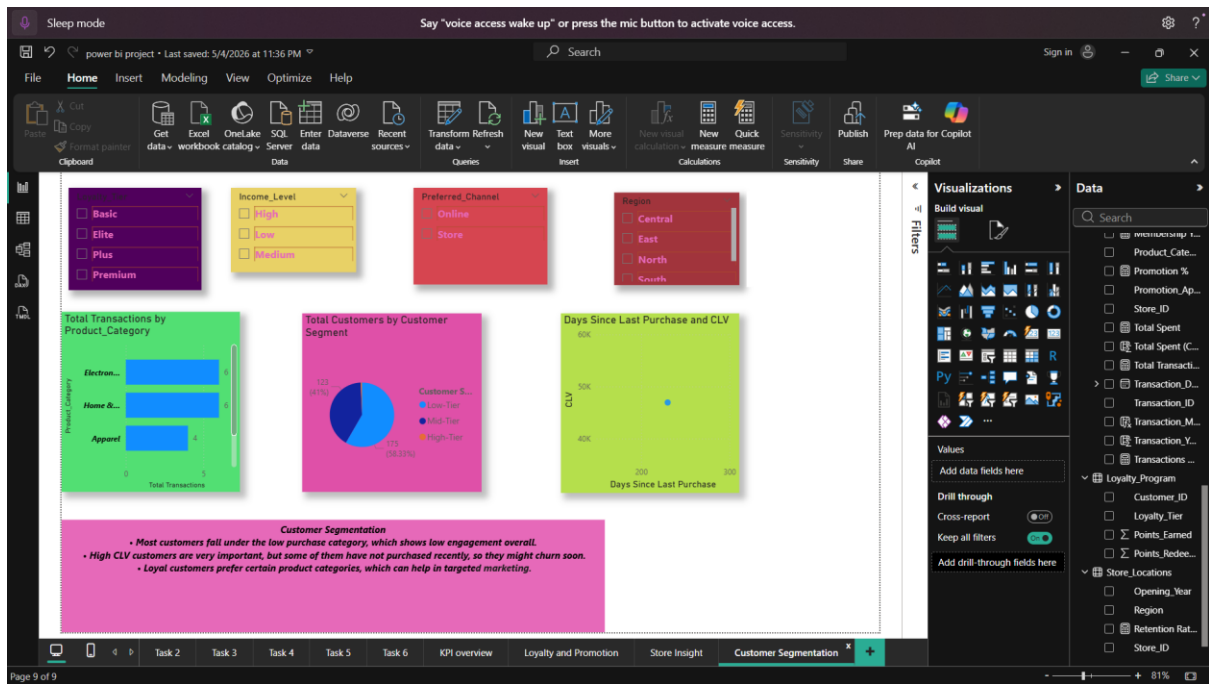
Page 2 Screenshot



Page 3 Screenshot



Page 4 Screenshot



Video Explanation Link

<https://www.loom.com/share/051068021bf74605a22813cb0a9704a1>